

|   |     |      |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                            |                        |
|---|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2 | (a) |      | The first law of thermodynamics states that the <u>increase in internal energy of a system</u> is the <u>sum of the heat (thermal energy) supplied to the system and the work done on the system.</u>                                      |                                                                                                                                                                                                                                                                                                                                                                                                            | B1                     |
|   | (b) | (i)  | 1.                                                                                                                                                                                                                                         | $W_{on} = -P\Delta V$ $= -(2.79 \times 10^5) [(1125 - 950) \times 10^{-6}]$ $= -48.825 = -48.8 \text{ J (3sf)}$ <p>Max 1m for positive answers.</p>                                                                                                                                                                                                                                                        | C1<br>A1               |
|   |     |      | 2.                                                                                                                                                                                                                                         | $\Delta U = Q_{in} + W_{on}$ $W_{on} = 0, \text{ since no volume change for } Q \rightarrow R$ $Q_{in} = \Delta U = \Delta \left( \frac{3}{2} pV \right)$ $= \frac{3}{2} [(2.10 \times 10^5)(1125 \times 10^{-6}) - (2.79 \times 10^5)(1125 \times 10^{-6})]$ $= -116.44$ $Q_{loss} = -Q_{in} = 116 \text{ J (3sf)}$ <p>Max 2m for negative answers, but do not penalize if same type of mistake in 1.</p> | C1<br><br>C1<br><br>A1 |
|   |     | (ii) | <p>Since there is no heat loss/gained, and <u>there is (positive) work done on the gas</u> through the compression (decrease in volume), the <u>internal energy will increase.</u></p> <p>Hence, <u>the temperature will increase.</u></p> |                                                                                                                                                                                                                                                                                                                                                                                                            | M1<br>A1               |

